

Professional Installation Manual

ABC Startup Solutions

Enterprise Ubuntu Server Deployment

Course: ITEP 414 – System Administration and Maintenance

Project: Week 3 Portfolio Project

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Server Hostname: server01

Operating System: Ubuntu Server LTS

Table of Contents

1. Introduction
 2. System Requirements
 3. Virtual Machine Configuration
 4. Ubuntu Server Installation
 5. Initial Server Configuration
 6. Server Verification
 7. Troubleshooting Guide
 8. BIOS vs UEFI
 9. Ubuntu Boot Process
 10. Windows Server Installation Summary
 11. Operating System Comparison
 12. Best Practices
 13. Lessons Learned
 14. References
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1. Introduction

This manual provides a complete procedure for deploying an Ubuntu Server virtual machine for ABC Startup Solutions. It is designed so that another Junior System Administrator can reproduce the installation using the documented configuration and procedures.

The server is intended to provide a foundation for future file sharing, remote administration, database hosting, web hosting, and internal services.

2. System Requirements

The required virtualization platform is Oracle VirtualBox or VMware Workstation.

The required operating system is Ubuntu Server LTS.

The virtual machine should have:

- 4 GB RAM
- 2 virtual processors
- 40 GB virtual disk
- NAT or Bridged networking
- Ubuntu Server LTS ISO

3. Virtual Machine Configuration

Component	Configuration
Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 processors
Storage	40 GB
Network	NAT
OS	Ubuntu Server LTS
Hostname	server01

4. Ubuntu Server Installation

The installation process consists of:

1. Booting from the Ubuntu Server ISO.
2. Selecting English.
3. Selecting the appropriate keyboard layout.
4. Configuring DHCP networking.
5. Recording the assigned IP address.
6. Setting the hostname to `server01`.
7. Creating a non-root administrative user.
8. Selecting Guided – Use Entire Disk.
9. Enabling OpenSSH Server.
10. Completing the installation.
11. Restarting the server.

5. Initial Server Configuration

After installation:

```
hostname
```

Then:

```
ip addr
```

Test connectivity:

```
ping -c 4 google.com
```

Update the system:

```
sudo apt update  
sudo apt upgrade -y
```

Verify SSH:

```
systemctl status ssh
```

The SSH service should report:

```
active (running)
```

6. Server Verification

The administrator should verify:

- Successful login
- Correct hostname
- Valid IP address
- Internet connectivity
- Successful system updates
- Running SSH service

Screenshots should be included as evidence for each verification task.

7. Troubleshooting Guide

Problem: No IP Address

Possible Cause

The virtual machine's network adapter may be disabled or incorrectly configured.

Solution

Check the VirtualBox/VMware network settings and confirm that the adapter is enabled and configured for NAT or Bridged mode as required.

Problem: Internet Ping Fails

Possible Cause

The server may not have network connectivity or DNS resolution may not be working.

Solution

Check:

```
ip addr
```

Then test network connectivity using the assigned gateway. Verify the virtual machine network configuration.

Problem: SSH Is Not Running

Possible Cause

OpenSSH Server may not have been installed or the service may not be running.

Solution

Check:

```
systemctl status ssh
```

If OpenSSH was not installed during setup, install it using the appropriate Ubuntu package-management procedure.

Problem: Hostname Is Incorrect

Solution

Verify:

```
hostname
```

Check the hostname configuration and ensure it is set to:

```
server01
```

8. BIOS vs UEFI

UEFI is the modern firmware standard used by most current computers. Compared with traditional BIOS, UEFI provides better support for modern storage technologies, GPT partitioning, modern boot management, and security features such as Secure Boot.

9. Ubuntu Boot Process

The Ubuntu boot process follows this general sequence:

```
Power On
  ↓
BIOS / UEFI Initialization
  ↓
Boot Device Detection
  ↓
GRUB Boot Loader
  ↓
Linux Kernel
  ↓
init / systemd
```

↓
Services Start
↓
Login Prompt

10. Windows Server Installation Summary

A separate Windows Server Evaluation virtual machine was created for comparison purposes.

The basic procedure included:

1. Creating a new virtual machine.
2. Attaching the Windows Server Evaluation ISO.
3. Installing Windows Server.
4. Assigning a computer name.
5. Creating an administrator password.
6. Completing installation.
7. Logging into the system.
8. Capturing screenshots as evidence.

11. Operating System Comparison

Windows Server is appropriate for Microsoft-focused enterprise environments. Ubuntu Server is a flexible open-source platform suitable for development, web hosting, cloud services, databases, and internal services. Rocky Linux provides an enterprise-oriented Linux platform suitable for organizations that want an Enterprise Linux-style environment.

12. Best Practices

Recommended server administration practices include:

- Keep the operating system updated.
- Use strong passwords.
- Avoid using the root account for normal administration.
- Use SSH securely.
- Restrict unnecessary services.
- Maintain regular backups.
- Monitor server performance.
- Document configuration changes.
- Use appropriate firewall rules.
- Protect server access.
- Keep installation media and documentation organized.

13. Lessons Learned

This project demonstrated that deploying a server requires careful planning, configuration, verification, security awareness, and documentation. Installing an operating system is only the first stage of server administration. A functioning server must also be tested, updated, secured, and documented before it can be considered ready for enterprise use.

14. References

The primary project instructions and requirements were based on the ITEP 414 System Administration and Maintenance Week 3 module provided by Laguna State Polytechnic University.